

Interactions

Last time: Class activity

$$\widehat{\text{Height}} = 40.689 - 7.728 \text{ IsMale} + 0.118 \text{ Age} \\ + 0.064 \text{ Age} \cdot \text{IsMale}$$

What is the estimated height for a girl aged 120 months (10 years)?

$$\text{IsMale} = 0$$

$$\text{Age} = 120$$

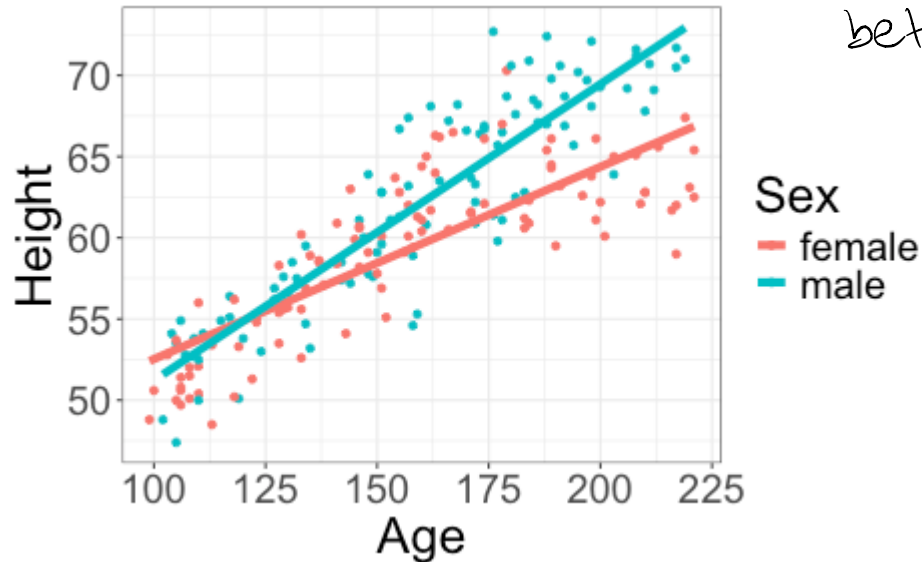
Last time: Class activity

$$\widehat{\text{Height}} = 40.689 - 7.728 \text{ IsMale} + 0.118 \text{ Age} \\ + 0.064 \text{ Age} \cdot \text{IsMale}$$

What is the estimated height for a girl aged 120 months (10 years)?

$$40.689 + 0.118(120) = 54.85 \text{ inches (about 4 feet 7 inches)}$$

Last time: Class activity



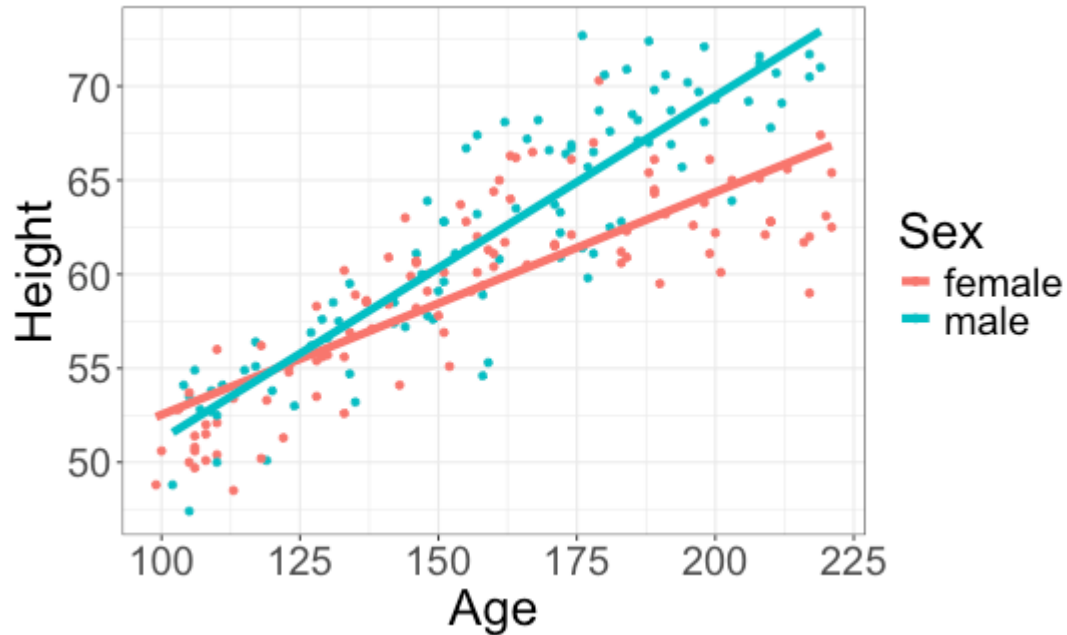
If we just want to describe the general relationship between age & height, then the interaction term doesn't add too much to the model

With interaction: $R^2_{adj} = 0.763$

Without interaction: $R^2_{adj} = 0.732$

Do you think an interaction is needed?

Last time: Class activity



I think an interaction is important

- + slopes look different
- + captures that boys and girls start off with similar heights, but boys eventually become taller.

Interactions with a categorical predictor

$$\begin{aligned}\text{bill length} = & \beta_0 + \beta_1 \text{IsChinstrap} + \beta_2 \text{IsGentoo} \\ & + \beta_3 \text{body mass} + \beta_4 \text{body mass} \cdot \text{IsChinstrap} \\ & + \beta_5 \text{body mass} \cdot \text{IsGentoo} + \varepsilon\end{aligned}$$

The slope on body mass depends on species:

- + Slope for Adelie penguins: β_3
- + Slope for Chinstrap penguins: $\beta_3 + \beta_4$
- + Slope for Gentoo penguins: $\beta_3 + \beta_5$

Interactions with a categorical predictor

$$\begin{aligned}\text{bill length} = & \beta_0 + \beta_1 \text{IsChinstrap} + \beta_2 \text{IsGentoo} \\ & + \beta_3 \text{body mass} + \beta_4 \text{body mass} \cdot \text{IsChinstrap} \\ & + \beta_5 \text{body mass} \cdot \text{IsGentoo} + \varepsilon\end{aligned}$$

The slope on body mass depends on species:

$$\text{slope} = \beta_3 + \beta_4 \text{IsChinstrap} + \beta_5 \text{IsGentoo}$$

We can do something similar for two quantitative variables

Interactions with two quantitative variables

Additive model (no interactions):

$$\text{bill length} = \beta_0 + \beta_1 \text{body mass} + \beta_2 \text{flipper length} + \varepsilon$$

+ Slope on body mass is β_1

+ Slope on body mass *does not depend on* flipper length

$$\text{bill length} = \beta_0 + \beta_1 \text{body mass} + \beta_2 \text{flipper length} + \beta_3 \text{body mass} \cdot \text{flipper length} + \varepsilon$$

for a fixed flipper length:

$$\dots \underbrace{(\beta_1 + \beta_3 \text{ flipper length})}_{\text{slope on body mass}} \text{body mass} \dots$$

(when flipper length is fixed)

$$\text{slope for body mass} = \beta_1 + \beta_3 \text{ flipper length}$$

Interactions with two quantitative variables

$$\text{bill length} = \beta_0 + (\text{slope}) \text{ body mass} + \beta_2 \text{flipper length} + \varepsilon$$

Making the slope depend on flipper length:

$$\text{slope on body mass} = \beta_1 + \beta_3 \text{flipper length}$$

Substituting:

$$\begin{aligned} \text{bill length} &= \beta_0 + (\beta_1 + \beta_3 \text{flipper length}) \text{ body mass} \\ &\quad + \beta_2 \text{flipper length} + \varepsilon \\ &= \beta_0 + \beta_1 \text{body mass} + \beta_2 \text{flipper length} \\ &\quad + \beta_3 \text{body mass} \cdot \text{flipper length} + \varepsilon \end{aligned}$$

Interactions with two quantitative variables

$$\begin{aligned}\text{bill length} = & \beta_0 + \beta_1 \text{body mass} + \beta_2 \text{flipper length} \\ & + \beta_3 \text{body mass} \cdot \text{flipper length} + \varepsilon\end{aligned}$$

- + body mass · flipper length is the *interaction* between body mass and flipper length
- + As before, it literally means multiplication

Interactions with two quantitative variables

$$\widehat{\text{bill length}} = -26.75 + 0.0059 \text{ body mass} + 0.339 \text{ flipper length} \\ - 0.000026 \text{ body mass} \cdot \text{flipper length}$$

What is the predicted bill length for a penguin with body mass 5000g and flipper length 200mm?

$$\begin{aligned} \widehat{\text{bill length}} &= -26.75 + 0.0059(5000) + \\ &\quad 0.339(200) \\ &\quad - 0.000026(5000)(200) \end{aligned}$$

Interactions with two quantitative variables

$$\widehat{\text{bill length}} = -26.75 + 0.0059 \text{ body mass} + 0.339 \text{ flipper length} \\ - 0.000026 \text{ body mass} \cdot \text{flipper length}$$

What is the predicted bill length for a penguin with body mass 5000g and flipper length 200mm?

$$-26.75 + 0.0059(5000) + 0.339(200) - 0.000026(5000)(200) \\ = 44.55$$

Interactions with two quantitative variables

$$\widehat{\text{bill length}} = -26.75 + 0.0059 \text{ body mass} + 0.339 \text{ flipper length} \\ - 0.000026 \text{ body mass} \cdot \text{flipper length}$$

What is the change in bill length associated with a 1g increase in body mass, for penguins with flipper length = 200mm?

terms involving body mass

$$0.0059 \text{ body mass} - 0.000026 \text{ body mass} \cdot \text{flipper length}$$

$$= \underbrace{(0.0059 - 0.000026 \text{ flipper length})}_{\text{slope on body mass, given flipper length}} \text{ body mass}$$

flipper length = 200mm:

$$0.0059 - 0.000026(200)$$

Interactions with two quantitative variables

$$\widehat{\text{bill length}} = -26.75 + 0.0059 \text{ body mass} + 0.339 \text{ flipper length} \\ - 0.000026 \text{ body mass} \cdot \text{flipper length}$$

What is the change in bill length associated with a 1g increase in body mass, for penguins with flipper length = 200mm?

$$0.0059 - 0.000026(200) = 0.0007$$

Interactions with two quantitative variables

$$\widehat{\text{bill length}} = -26.75 + 0.0059 \text{ body mass} + 0.339 \text{ flipper length} \\ - 0.000026 \text{ body mass} \cdot \text{flipper length}$$

Is the slope on body mass higher or lower for penguins with longer flippers?

Interactions with two quantitative variables

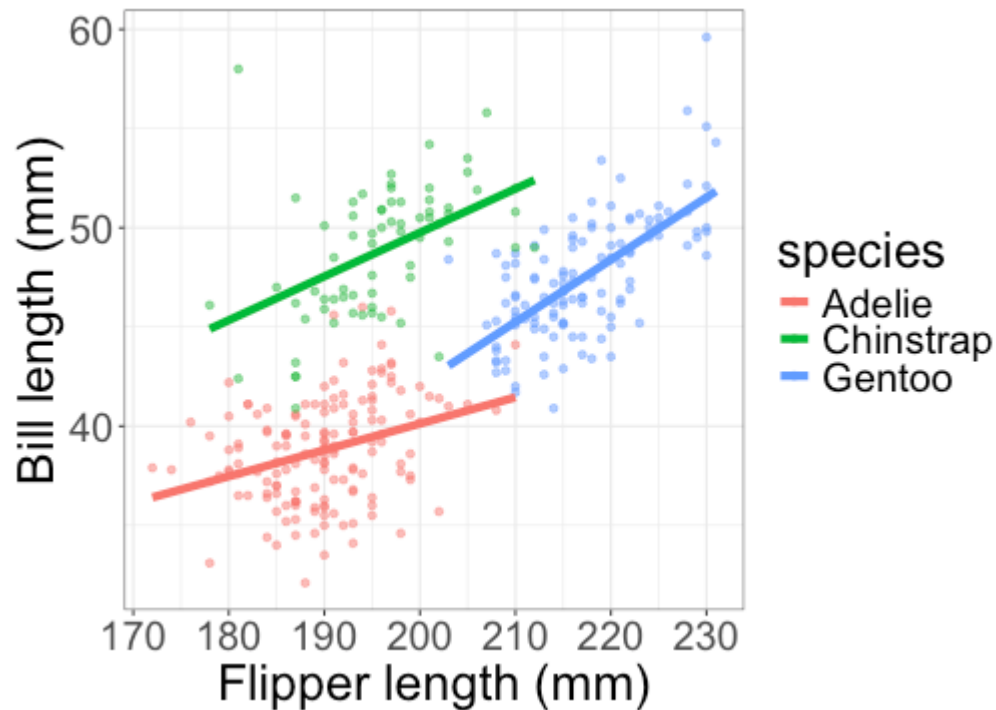
$$\widehat{\text{bill length}} = -26.75 + 0.0059 \text{ body mass} + 0.339 \text{ flipper length} \\ - 0.000026 \text{ body mass} \cdot \text{flipper length}$$

Is the slope on body mass higher or lower for penguins with longer flippers?

Lower, because the coefficient -0.000026 is negative.

Visualizing interactions

With a categorical variable, we can fit a different slope for each group:

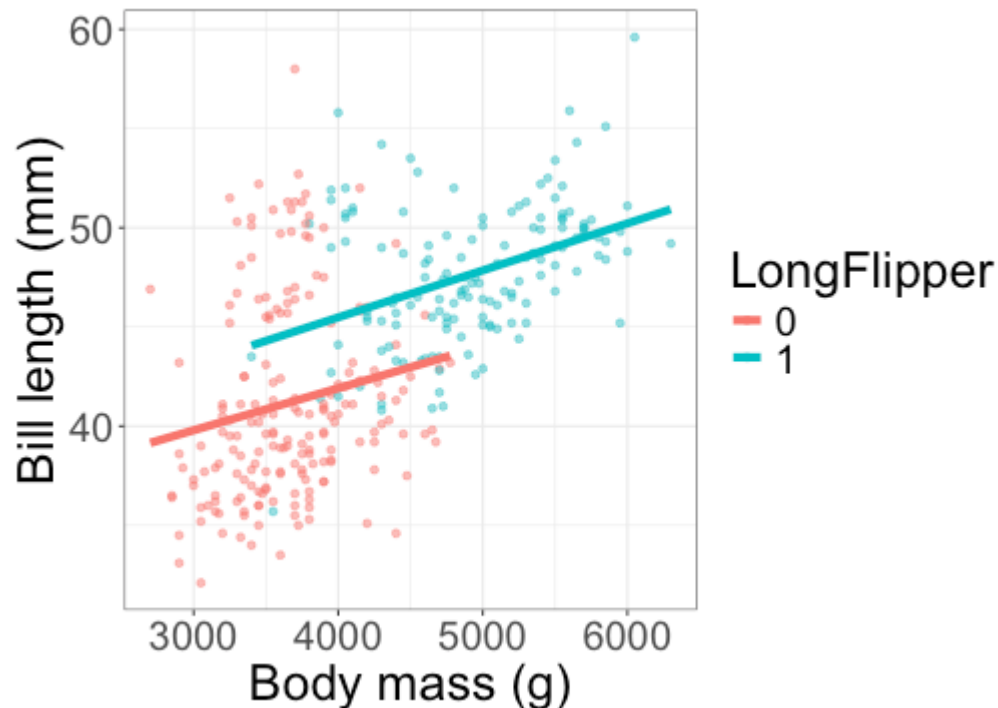


How would this work with two quantitative variables?

Visualizing interactions

Binarize:

$$LongFlipper = \begin{cases} 1 & \text{flipper length} > 200 \\ 0 & \text{flipper length} \leq 200 \end{cases}$$



Homework work time

Homework 6 involves a research question with an interaction between two quantitative variables. Spend the remainder of the class period getting started on HW 6.